An epigenome-wide association study of waist circumference in Chinese monozygotic twins.

PMID: 38773251 | Indexed in PubMed

Central obesity poses significant health risks because it increases susceptibility to multiple chronic diseases. Epigenetic features such as DNA methylation may be associated with specific obesity traits, which could help us understand how genetic and environmental factors interact to influence the development of obesity. This study aims to identify DNA methylation sites associated with the waist circumference (WC) in Northern Han Chinese population, and to elucidate potential causal relationships. A total of 59 pairs of WC discordant monozygotic twins (ΔWC >0) were selected from the Qingdao Twin Registry in China. Generalized estimated equation model was employed to estimate the methylation levels of CpG sites on WC. Causal relationships between methylation and WC were assessed through the examination of family confounding factors using FAmiliaL CONfounding (ICE FALCON). Additionally, the findings of the epigenome-wide analysis were corroborated in the validation stage. We identified 26 CpG sites with differential methylation reached false discovery rate (FDR) < 0.05 and 22 differentially methylated regions (slk-corrected $p < 0.05$) strongly linked to WC. These findings provided annotations for 26 genes, with notable emphasis on MMP17, ITGA11, COL23A1, TFPI, A2ML1-AS1, MRGPRE, C2orf82, and NINJ2. ICE FALCON analysis indicated the DNA methylation of ITGA11 and TFPI had a causal effect on WC and vice versa ($p < 0.05$). Subsequent validation analysis successfully replicated 10 ($p < 0.05$) out of the 26 identified sites. Our research has ascertained an association between specific epigenetic variations and WC in the Northern Han Chinese population. These DNA methylation features can offer fresh insights into the epigenetic regulation of obesity and WC as well as hints to plausible biological mechanisms.

Heritability of Body Mass Index Among Familial Generations.

PMID: 38941093 | Indexed in PubMed

Studies on the familial effects of body mass index (BMI) status have yielded a wide range of data on its heritability. To assess the heritability of obesity by measuring the association between the BMIs of fathers, mothers, and their offspring at the same age. This cohort study used data from population-wide mandatory medical screening before compulsory military service in Israel. The study included participants examined between January 1, 1986, and December 31, 2018, whose both parents had their BMI measurement taken at their own prerecruitment evaluation in the past. Data analysis was performed from May to December 2023. Spearman correlation coefficients were calculated for offsprings' BMI and their mothers', fathers', and midparental BMI percentile (the mean of the mothers' and fathers' BMI cohort- and sex-specific BMI percentile) to estimate heritability. Logistic regression models were applied to estimate the odds ratios (ORs) and 95% CIs of obesity compared with healthy BMI, according to parental BMI status. A total of 447 883 offspring (235 105 male [52.5%]; mean [SD] age, 17.09 [0.34] years) with both parents enrolled and measured for BMI at 17 years of age were enrolled in the study, yielding a total study population of 1 343 649 individuals. Overall, the correlation between midparental BMI percentile at 17 years of age and the offspring's BMI at 17 years of age was moderate ($\rho = 0.386$). Among female offspring, maternal-offspring BMI correlation ($\rho = 0.329$) was somewhat higher than the paternal-offspring BMI correlation ($\rho = 0.266$). Among trios in which both parents had a healthy BMI, the prevalence of overweight or obesity in offspring was 15.4%; this proportion increased to 76.6% when both parents had obesity and decreased to 3.3% when both parents had severe underweight. Compared with healthy weight, maternal (OR, 4.96; 95% CI, 4.63-5.32), paternal (OR, 4.48; 95% CI, 4.26-4.72), and parental (OR, 6.44; 95% CI, 6.22-6.67) obesity (midparent BMI in the ≥95th percentile) at 17 years of age were associated with increased odds of obesity among offspring. In this cohort study of military enrollees whose parents also underwent prerecruitment evaluations, the observed correlation between midparental and offspring BMI, coupled with a calculated narrow-sense heritability of 39%, suggested a substantive contribution of genetic factors to BMI variation at 17 years of age.

[Familial obesity as a predictor of child obesity].

PMID: 16526252 | Indexed in PubMed

Studies carried out in various parts of the world indicate that family obesity significantly affects the incidence of obesity in children. This is especially a characteristic of children whose both parents are obese. The study was conducted using a polling method. Questionnaires were filled out by parents and brothers and sisters, including their body height and weight. The collected data served as the basis for assessing the family nutritional status. Statistical analysis of the results showed that obese children frequently have obese parents, brothers and sisters in regard to normal-weight children. Differences are statistically significant in relation to fathers ($p=0.043$), i.e. statistically obese schoolchildren have more frequently obese fathers than those of normal nutritional status. Other differences could not be considered significant ($p > 0.05$). Obese children have more often obese parents, brothers and sisters than normal-weight children. It was found that the nutritional status of moderately and extremely obese children was quite different from that of normal-weight children and that there was a statistically significant dependence between the nutritional status of children and their fathers. This research showed that family obesity is a potential contributing factor to obesity of schoolchildren.

Familial aggregation of morbid obesity.

PMID: 16353356 | Indexed in PubMed

Recent studies have shown major gene effects for obesity in randomly ascertained families. To investigate the familial aggregation of a specific subset of obesity, which is particularly prone to medical complications, families with morbid obesity were studied. This condition occurs in 1%-2% of the population and is defined as 45.5 kg (100 pounds) or more over ideal weight. First-degree relatives of 221 morbidly obese probands (1560 adults) were identified, and height and weight (current and greatest) were obtained from each family member. Morbid obesity occurred in the family members of the probands 8 times more often than in the general population. Of the morbidly obese probands, 48% had one or more first-degree relatives who were also morbidly obese compared to a 6% population estimate. By the ages of 20-24, 12% of the morbidly obese probands were already 45.5 kg or more overweight, and 45% were 22.7 kg (50 pounds) or more overweight. There was little difference in the prevalence of familial morbid obesity by the gender of the probands: 47% of the male probands and 48% of the female probands had another morbidly obese relative, while 67% and 53% of the early onset (before age 25) male and female probands, respectively, had one or more first-degree relatives who were also morbidly obese. In addition to the extreme degree of familial aggregation, the prevalence of morbid obesity in parent-offspring sets was calculated within the morbidly obese families. Morbidly obese families who have one or two morbidly obese parents have a 2.6 times increased risk ($p<0.002$) of having one or more morbidly obese adult offspring, compared to families who have neither parent morbidly obese. Evidence for trimodality of the body mass index distribution was found for each gender ($p = 0.0006$ for male relatives and p = 0.075 for female relatives). The strong familial aggregation of morbid obesity indicates the need for further understanding of the genetic determinants of this extreme clinical disorder and how environmental factors affect the genetic expression of the trait.

Etiology of massive obesity: role of genetic factors.

PMID: 9717415 | Indexed in PubMed

It is well established that there is a significant familial aggregation of obesity, although most of the evidence regarding the genetic basis of obesity has been derived from overweight and moderately obese cases. Less is known about the contribution of genetic factors in severely obese individuals. This paper reviews the available evidence regarding the extent of familial aggregation of morbid obesity and the contribution of specific genes. The results of available studies suggest a stronger degree of familial resemblance for morbid obesity [body mass index (BMI > 40 kg/m2)] than for more moderate levels of obesity (BMI < 40 kg/m2). Evidence from human association and linkage studies, performed with markers surrounding human homologs of the genes involved in mouse models of obesity, revealed that these genes tend to be linked more often to severe obesity than to moderate levels of obesity.

Genetics of obesity.

PMID: 28697287 | Indexed in PubMed

The prevalence and cost of overweight and obesity are projected to continue to rise in our current sedentary, energy-rich environment. The heritability of obesity is reported to be 40-80%. The objectives of this article are to review recent genetic discoveries about the basis of human obesity; describe familial or syndromic obesity, which is rare but presents early and should, if suspected, be referred for full specialist diagnosis of the underlying genetic disorder; and summarise immediate implications for general practice. Increasingly, specialised genetic studies are helping to unravel the complex physiology underlying the regulation and control of body fatness. Rare but serious syndromic obesity can now be identified early in life. Possible targets for future drug treatment are currently being found and tested, offering hope for the future. Understanding the heritable nature of obesity should also help to relieve the blame and guilt from the professional's approach to current weight management.

Dietary patterns and associated lifestyles in individuals with and without familial history of obesity: a cross-sectional study.

PMID: 17076904 | Indexed in PubMed

Familial history of obesity (FHO) and certain dietary habits are risk factors for obesity. The objectives of this cross-sectional study were 1) to derive dietary patterns using factor analysis in a population of men and women with and without FHO; 2) to compare mean factor scores for each dietary pattern between individuals with and without FHO; and 3) to examine the association between these patterns and anthropometric, lifestyle and sociodemographic variables. A total of 197 women and 129 men with a body mass index <30 kg/m2 were recruited. A positive FHO (FHO+) was defined as having at least one obese first-degree relative and a negative FHO (FHO-) as no obese first-degree relative. Dietary data were collected from a food frequency questionnaire. Factor analysis was performed to derive dietary patterns. Mean factor scores were compared using general linear model among men and women according to FHO. Regression analyses were performed to study the relationship between anthropometric, lifestyle and sociodemographic variables, and each dietary pattern. Two dietary patterns were identified in both men and women : the Western pattern characterized by a higher consumption of red meats, poultry, processed meats, refined grains as well as desserts, and the Prudent pattern characterized by greater intakes of vegetables, fruits, non-hydrogenated fat, and fish and seafood. Similar Western and Prudent factor scores were observed in individual with and without FHO. In men with FHO+, the Western pattern is negatively associated with age and positively associated with physical activity, smoking, and personal income. In women with FHO-, the Prudent pattern is negatively associated with BMI and smoking and these pattern is positively associated with age and physical activity. Two dietary patterns have been identified among men and women with and without FHO. Although that FHO does not seem to influence the adherence to dietary patterns, results of this study suggest that anthropometric, lifestyle and sociodemographic variables associated with dietary patterns differ according to FHO and gender.

Genetic obesity: an update with emerging therapeutic approaches.

PMID: 36203267 | Indexed in PubMed

Based on the genetic contribution, childhood obesity can be classified into 3 groups: common polygenic obesity, syndromic obesity, and monogenic obesity. More genetic causes of obesity are being identified along with the advances in the genetic testing. Genetic obesities including syndromic and monogenic obesity should be suspected and evaluated in children with early-onset morbid obesity and hyperphagia under 5 years of age. Patients with syndromic obesity have early-onset severe obesity associated specific genetic syndromes including Prader-Willi syndrome, Bardet-Biedle syndrome, and Alstrom syndrome. Syndromic obesity is often accompanied with neurodevelopmental delay or dysmorphic features. Nonsyndromic monogenic obesity is caused by variants in single gene which are usually involved in the regulation of hunger and satiety associated with the hypothalamic leptin-melanocortin pathway in central nervous system. Unlike syndromic obesity, patients with monogenic obesity usually show normal neurodevelopment. They would be presented with hyperphagia and early-onset severe obesity with additional clinical symptoms including short stature, red hair, adrenal insufficiency, hypothyroidism, hypogonadism, pituitary insufficiencies, diabetes insipidus, increased predisposition to infection or intractable recurrent diarrhea. Identifying patients with genetic obesity is critical as new innovative therapies including melanocortin 4 receptor agonist have become available. Early genetic evaluation enables to identify treatable obesity and provide timely intervention which may eventually achieve favorable outcome by establishing personalized management.

Pre and post-natal risk and determination of factors for child obesity.

PMID: 27928443 | Indexed in PubMed

Obesity is considered a condition presenting a complex, multi-factorial etiology that implies genetic and non-genetic factors. The way the available information should be efficiently and strategically used in the obesity and overweight prohylaxisprogrammes for children all over the world is still unclear for most of the risk factors. Mothers' pre-conception weight and weight gain during pregnancy are two of the most important prenatal determinants of childhood obesity. Maternal obesity and gestational weight gain are associated with foetal macrosomia and childhood obesity, and this effect extends into adulthood. Obesity and the metabolic syndrome in children originate in intrauterine life. The current obesity epidemic is probably the result of our evolutive inheritance associated with the consumption of highly processed food with an increased calorific value. The determination of risk factors involved in child obesity are: genetic predisposition, diet, sedentary behaviors, socioeconomic position, ethnic origin, microbiota, iatrogenic, endocrine diseases, congenital and acquired hypothalamic defects, usage of medications affecting appetite. However, the vast majority of patients will not have any of these identifiable conditions. Regardless of the aetiology, all the patients should be considered for modifiable lifestyle risk factors and screened for the complications of obesity.

A glyco-engineering approach for site-specific conjugation to Fab glycans.

PMID: 36447399 | Indexed in PubMed

Effective processes for synthesizing antibody-drug conjugates (ADCs) require: 1) site-specific incorporation of the payload to avoid interference with binding to the target epitope, 2) optimal drug/antibody ratio to achieve sufficient potency while avoiding aggregation or solubility problems, and 3) a homogeneous product to facilitate approval by regulatory agencies. In conventional ADCs, the drug molecules are chemically attached randomly to antibody surface residues (typically Lys or Cys), which can interfere with epitope binding and targeting, and lead to overall product heterogeneity, long-term colloidal instability and unfavorable pharmacokinetics. Here, we present a more controlled process for generating ADCs where drug is specifically conjugated to only Fab N-linked glycans in a narrow ratio range through functionalized sialic acids. Using a bacterial sialytransferase, we incorporated N-azidoacetylneuraminic acid (Neu5NAz) into the Fab glycan of cetuximab. Since only about 20% of human IgG1 have a Fab glycan, we extended the application of this approach by using molecular modeling to introduce N-glycosylation sites in the Fab constant region of other therapeutic monoclonal antibodies. We used trastuzumab as a model for the incorporation of Neu5NAz in the novel Fab glycans that we designed. ADCs were generated by clicking the incorporated Neu5NAz with monomethyl auristatin E (MMAE) attached to a self-immolative linker terminated with dibenzocyclooctyne (DBCO). Through this process, we obtained cetuximab-MMAE and trastuzumab-MMAE with drug/antibody ratios in the range of 1.3 to 2.5. We confirmed that these ADCs still bind their targets efficiently and are as potent in cytotoxicity assays as control ADCs obtained by standard conjugation protocols. The site-directed conjugation to Fab glycans has the additional benefit of avoiding potential interference with effector functions that depend on Fc glycan structure.